

MAT 320

Quiz 2

Fall 2025

1. If the complex number $-\frac{1}{2} + \frac{\sqrt{3}}{2}i$ is rotated by $-\pi/6$ radians (clockwise), what is the resulting complex number?
- a) $1 - i$ b) $\frac{1}{2} - \frac{\sqrt{3}}{2}i$ c) $\frac{1}{2} + \frac{\sqrt{3}}{2}i$ d) -1 e) i

Correct Answer: i

2. Simplify: $e^{i3t} - e^{-i3t}$
- a) $2\cos(3t)$ b) $2i\sin(3t)$ c) $2\cos(3t) + 2i\sin(3t)$ d) $\cos(3t) + i\sin(3t)$ e) 1

Correct Answer: $2i\sin(3t)$

3. Factor out from the following expression an exponential function of t .

$$2e^{i\frac{\pi}{4}t} + e^{i\frac{\pi}{4}(t-2)}$$

What is the remaining factor?

- a) $1 - i$ b) $1 + 2i$ c) $1 + i$ d) $2 - i$ e) $1 + 2i$

Correct Answer: $2 - i$

4. The function $f(z) = \alpha z$ rotates its input by some angle, and also scales by a factor of $\sqrt{5}$ for which complex constant α ?

- a) $-2 + i$ b) $1 - i$ c) $3 + 2i$ d) $\sqrt{5}(1 + i)$ e) $2 + 2i$

Correct Answer: $-2 + i$

5. Consider the digital filter F_1 : $y_t = x_t - \frac{1}{4}x_{t-1}$. What is the transfer function $\mathcal{H}(z)$ for this filter?

- a) $1 - \frac{1}{4}z$ b) $\frac{z}{z-\frac{1}{4}}$ c) $\frac{1}{1-\frac{1}{4}z}$ d) $1 - \frac{1}{4}z^{-1}$ e) $1 - 4z$

Correct Answer: $1 - \frac{1}{4}z^{-1}$

6. Same filter F_1 as in the previous question. The transfer function of this filter has a zero at z equal to:

- a) $-\frac{1}{4}$ b) $\frac{1}{4}$ c) $\frac{1}{4}(1 + i)$ d) $-\frac{1}{4}(1 + i)$ e) 0

Correct Answer: $\frac{1}{4}$

7. Same filter F_1 as in the previous question. The transfer function of this filter has a pole at z equal to:

- a) $-\frac{1}{4}$ b) $\frac{1}{4}$ c) $\frac{1}{4}(1 + i)$ d) $-\frac{1}{4}(1 + i)$ e) 0

Correct Answer: 0

8. Same filter F_1 as in the previous question. What is the frequency response $H(\omega)$ of this filter for $\omega = 0$?

- a) $\frac{3}{4}$ b) $\frac{4}{3}$ c) 1 d) $\frac{\sqrt{3}}{5}(1 + i)$ e) $\frac{\sqrt{2}}{5}(1 + i)$

Correct Answer: $\frac{3}{4}$

9. Same filter F_1 as in the previous question. What is the magnitude response $|H(\omega)|$ of this filter for $\omega = \pi/2$?

- a) $\sqrt{\frac{17}{18}}$ b) $\sqrt{\frac{12}{17}}$ c) $\frac{\sqrt{17}}{4}$ d) $\frac{\sqrt{17}}{5}$ e) $\frac{\sqrt{5}}{17}$

Correct Answer: $\frac{\sqrt{17}}{4}$

10. Suppose we use sampling rate $f_s = 44100$ Hz. What is the frequency of smallest positive value which is an alias of the frequency 32100 Hz, but is not equal to this frequency?

- a) 12050 b) 12000 c) 12500 d) 10050 e) 6000

Correct Answer: 12000